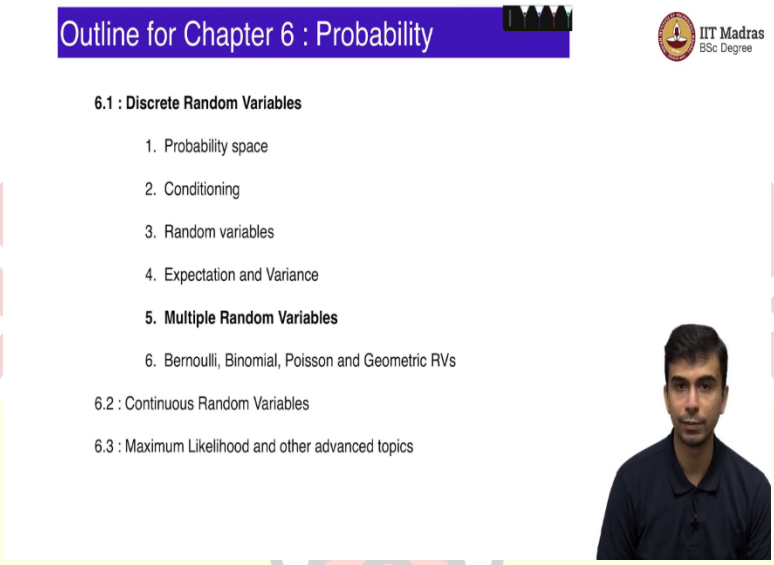


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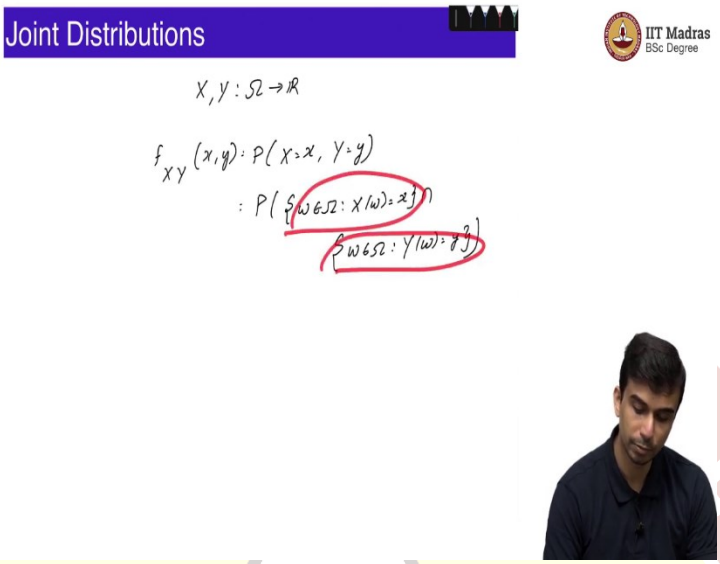
Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras
Multiple Random Variables

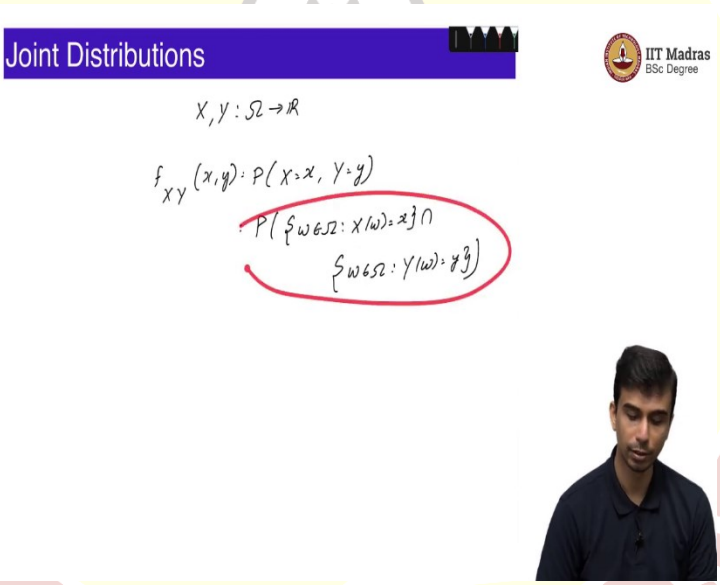
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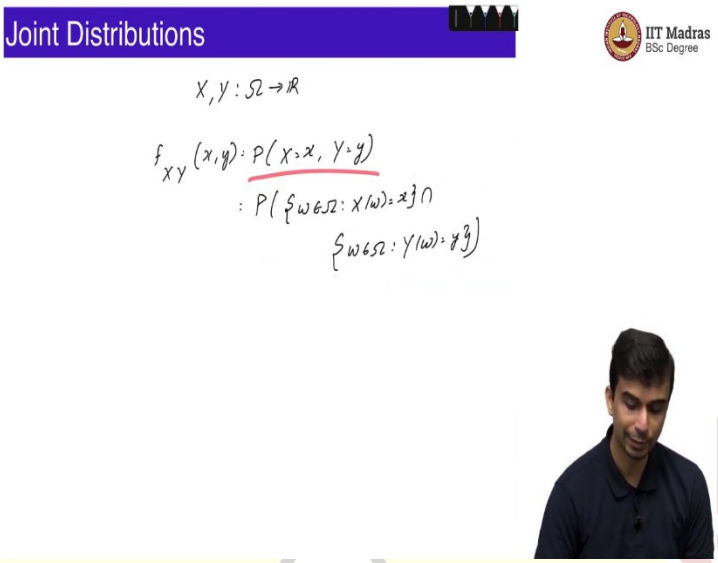


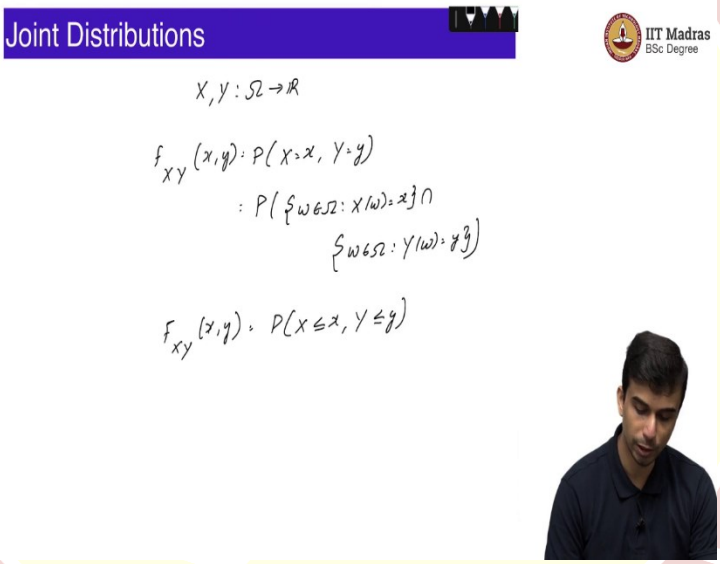
Until now, we have been dealing with individual single random variables, and their properties, such as their probability mass function, their cumulative distribution function, expectation, variance and so on. However, in several practical problems what you will be faced with is you want to analyze multiple random variables simultaneously. What are the key tools for doing that? We will look at that in this lecture.

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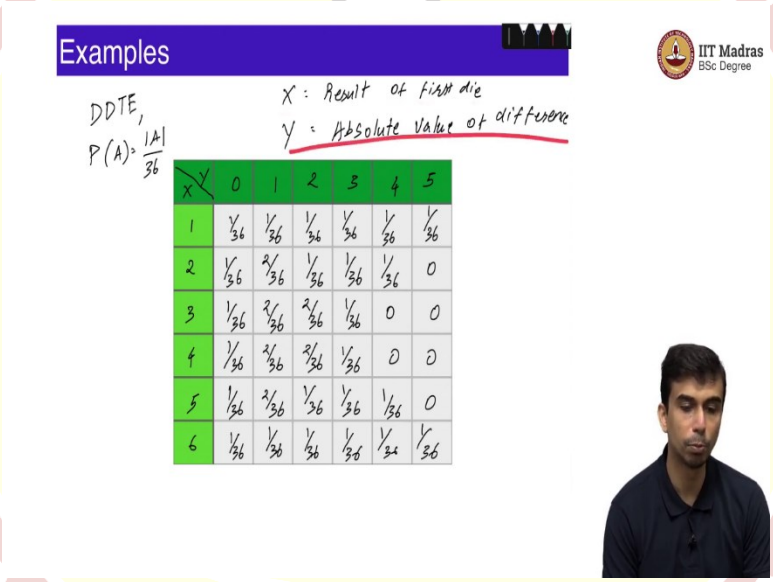
The first tool is the most basic tool, which is we generalize the probability mass function of individual random variables to multiple random variables. Let us take the simple case of 2 random variables, let us say x and y are 2 random variables from ω to R. The joint probability mass function of this pair of random variables x and y is denoted by small f subscript x y and f x y of x , y is simply the probability of x equal to x and y equal to y.

In, in terms of events this means the probability of the set ω in the sample space such that x of ω equal to x and which is denoted by intersection set of ω in ω such that y of ω equal to y.

So as this is a valid event, and this is a valid event so intersection of events is also valid. And so, the probability of, this is a valid probability measure. So, that is what we mean by probability of x equal to x and probability of y equal to y. This generalizes the probability mass function to multiple random variables. Easily it also generalizes to 3 or 4 or 5 or any number of random variables.

Similarly, you can also have a cumulative distribution function, which is capital F x y of x , y, which is simply the probability of x less than or equal to x and y less than or equal to y. So, this generalizes the cumulative distribution function of single random variables to multiple random variables.

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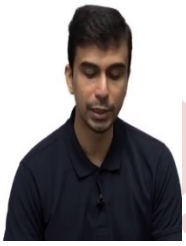
Examples

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DDTE,
$P(A) = \frac{|A|}{36}$

X : Result of first die
Y : Absolute value of difference

X \ Y	0	1	2	3	4	5
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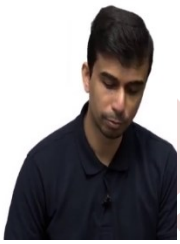
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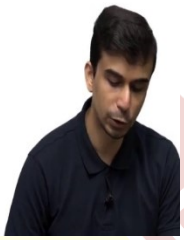
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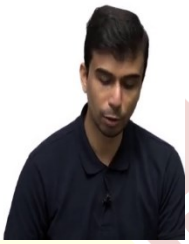
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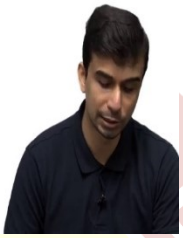
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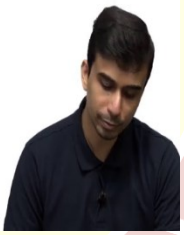
Examples

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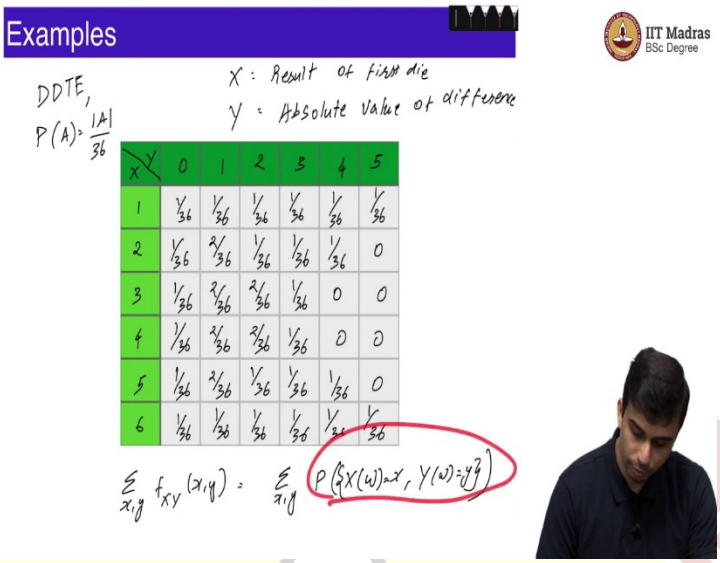
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Let us take a look at a simple example to make these ideas concrete. Let us consider the standard double dice throw experiment, with the counting probability measure which corresponds to using both the dies being fair.

Let us take the 2 random variables. X being the result of the first dice, and Y being the absolute value of the difference between the 2 dice. We have individually analyzed both these random variables and we know their probability mass function, expectation and so on. But we have not analyzed them together till now. Let us do that.

We know that the result of the first die can take values between 1 to 6, and the absolute value of the difference can take values between 0 to 5. So, the probability mass function, to know the probability mass function you need to know this entire table. So, what, how to interpret this table, for example, this number here corresponds to the probability of x equal to 1, and y equal to 0.

So, what is that? Well, x equal to 1 means the result of the first dice is, first dice is 1, and the absolute value 0 means that the second dice also has to be 1. So, that is exactly equal to 1 by 36. 1 , 1 is the only event that, only outcome that results in this. So, that is 1, 1 by 36. Similarly, you can fill in the entire first row would be 1 by 36 all throughout.

The second row is a little bit more interesting. Let us say the first die roll is 2, what is the probability the first die roll is 2 and the difference is 0. So, the only outcome for this is 2 , 2, which is 1 by 36. But the absolute difference being 1 means that both 2 , 1 and 2 , 3 are

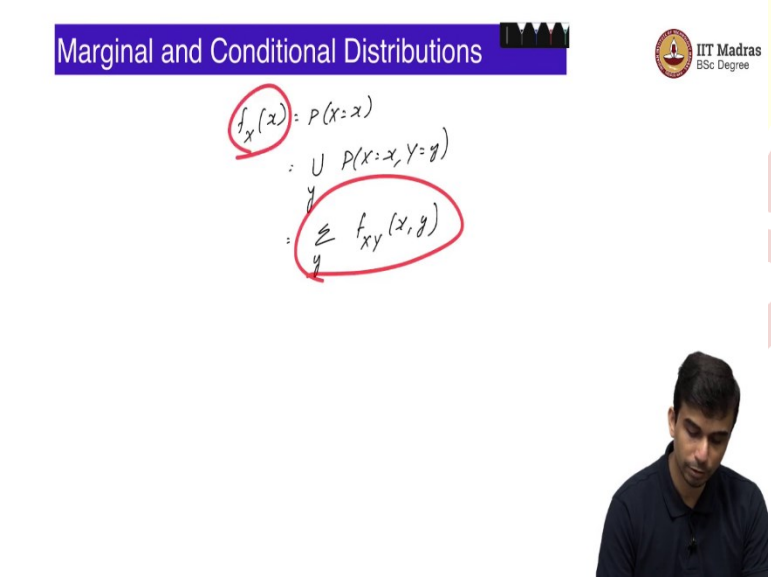
valid outcomes which result in this particular configuration of x being 2 and y being 1. So, the probability of this event is 2 by 36.

And similarly, you can fill in this entire row, here. Note that probability of x equal to 2 and y equal to 5 is 0. If the first die roll is 2, there is no way the difference can be 5. Similarly, you can fill in rows 3, 4, 5 and 6 and so on. And a key thing to note is if you added the entire table that should exactly correspond to 1.

Why is that? Well for the exact same reason this probability mass function should sum to 1, the prob, the joint probability mass function should also sum to 1. Why is that? Well sum over x , y f x y of x , y is exactly equal to pro, sum over x , y probability of x of ω equal to x and y of ω equal to y.

These sets are disjoint and their union corresponds to the entire sample space, so which is equal to probability of ω, which is equal to 1. So, that is the reason why this entire table should sum to 1, and as a quick check you can see that that does indeed happen. So, that, this is a pretty simple example for doing joint distributions. This is called a joint mass function or a joint distribution.

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Marginal and Conditional Distributions

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$$f_X(x) = P(X = x)$$
$$= \sum_y P(X = x, Y = y)$$
$$= \sum_y f_{XY}(x, y)$$
$$f_{Y|X}(y|x) = \sum_{x \in Range(X)} f_{XY}(x, y)$$



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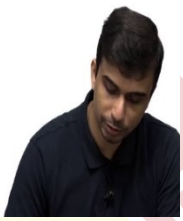


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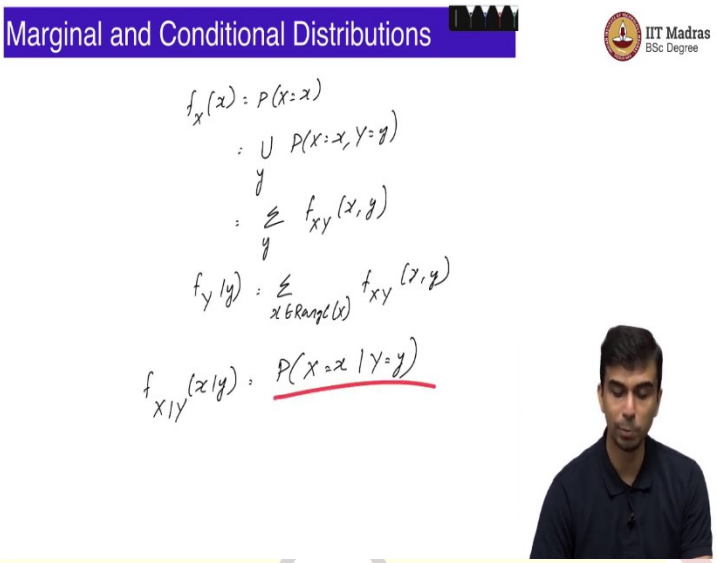
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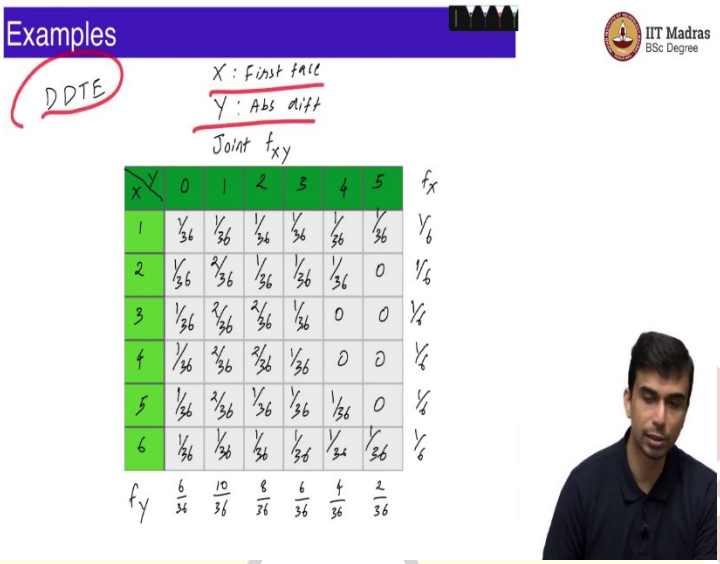
So, there are a couple of derived distributions from the joint distribution which are the marginal and the conditional. The marginal is something which you have already seen, which is simply the pmf of the individual random variables. What is f x? What can you say about f x of x? Well f x of x is simply the probability of x equal to x, which is simply the union over all y probability of x equal to x and y equal to y.

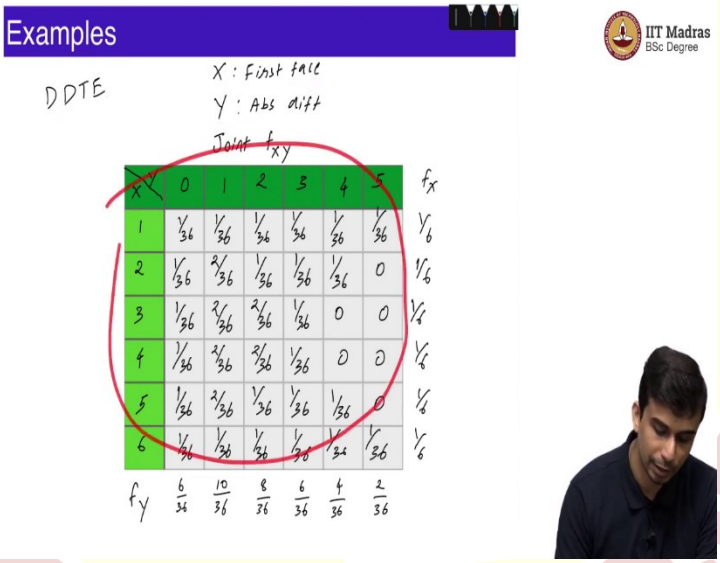
This is equal to sum over y in range of y, f x y of x , y. So, you sum over y, the, you sum the joint over y, you get the marginal over x. Similarly, you can have f y of y is sum over x in range of x f x y of x , y. By marginalizing or summing out over x you get the marginal distribution over y, and similarly by summing over y you get the marginal distribution over x. This is pretty simple and straight forward.

Then you have a couple of other derived distributions, which are the conditional distributions. f x given y of x given y is simply the probability of x equal to x given y equal to y. x equal to x is an event, and y equal to is also an event. So, this is a valid probability statement. So, you can have the left hand, both these are events so we know, already know conditional probability of events. This is the definition.

But we can go further and say probability of x equal to x given y equal to y is simply probability of x equal to x and y equal to y by probability of y equal to y, which is exactly equal to f x y of x , y by f y of y. Similarly, you can have f y given x of y given x which is f x y of x , y by f x of x. We will illustrate these distributions also for the same example we had.

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Examples

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DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

IIT Madras BSc Degree

DDTE

X : First face

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
2	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
3	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
5	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
6	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
f_Y	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	



Examples

IIT Madras BSc Degree

DDTE

X : First face

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
2	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
3	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
5	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
6	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
f_Y	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$	



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Examples

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X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

IIT Madras BSc Degree

DDTE

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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Y : Abs diff

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X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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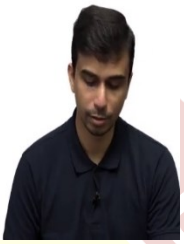
DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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DDTE

X : First face

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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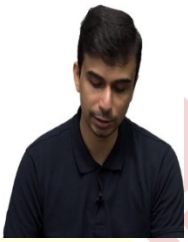
DDTE

X : First face

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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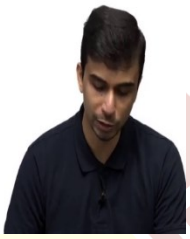
DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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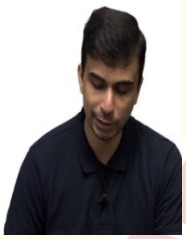
DDTE

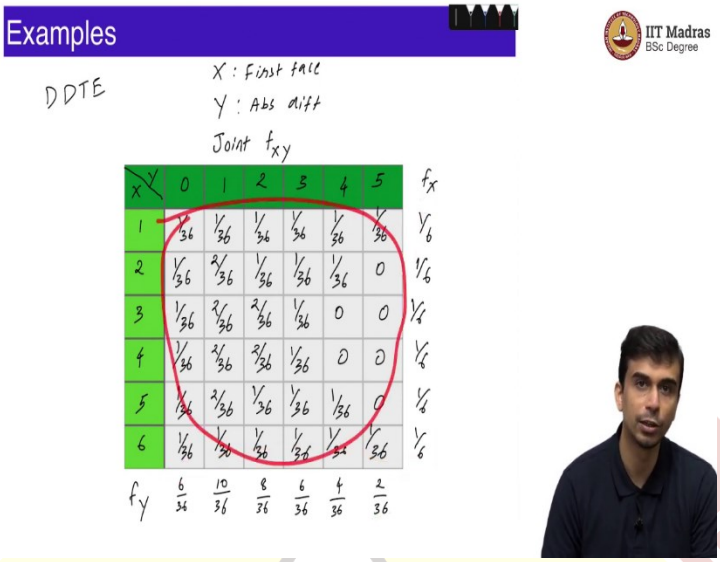
X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	





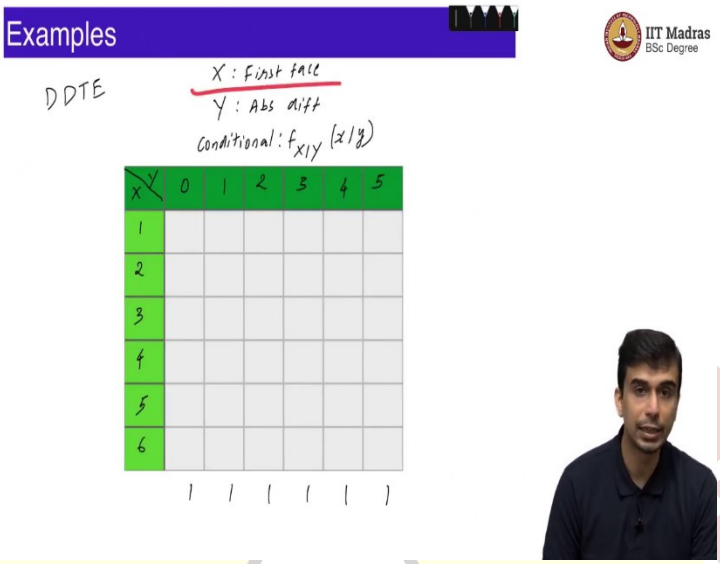
Let us take the double dice throw experiment with x being the first face, and y being the absolute difference. We already have the marginal distribution. We can compute, sorry, we already have the joint distribution. We can compute the marginals just by summing over the rows or columns. To get the marginal distribution f x we just sum over the columns, and you get each row sum corresponds to f x.

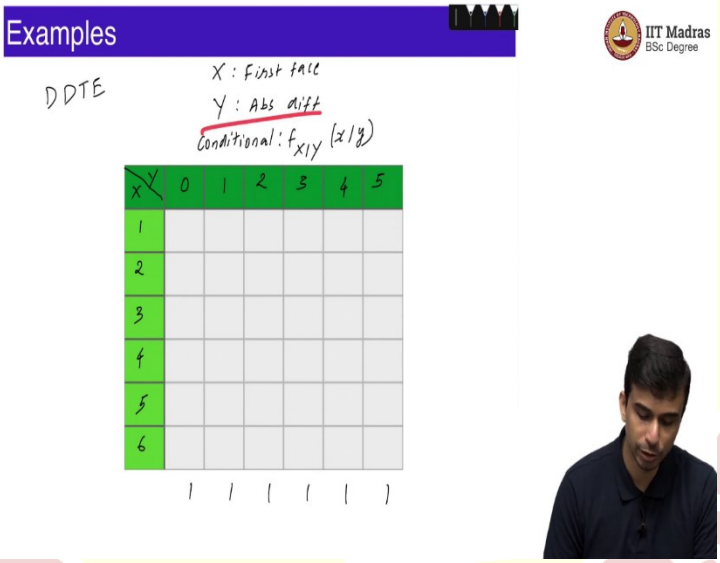
And the sum over the first row you get 1 by 6. So, move to second row, you get 1 by 6 and so on. And you can see that f x ends up being 1 by 6, which you already know. The probability mass function on the first face is exactly equal to this, you know that.

And by summing over the column, each column sum corresponds to a marginal distribution f y of y. So, something over the first column corresponds to the probability of y equal to y which is 6 by 36, or for probability y equal to 0, which is 6 by 36 and similarly something about the second column gives you probability of y equal to 1, which is 10 by 36 and so on.

This is also known when we computed the probability mass function of the random variable y previously. And you can see this matches the previously computed probability mass function of y. So, this is the marginal f x and this is the marginal f y, and the joint f x y.

(Refer Slide Time: 09:10)





Examples

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DDTE

X : First face

Y : Abs diff

Conditional : $f_{X|Y}(x|y)$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						

1 1 1 1 1 1



Examples

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DDTE

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Conditional : $f_{X|Y}(x|y)$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						

1 1 1 1 1 1



Examples

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X : First face

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Conditional : $f_{X|Y}(x|y)$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						

1 1 1 1 1 1



Examples

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DDTE

X : First face

Y : Abs diff

Conditional : $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						

1 1 1 1 1 1



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Examples

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DDTE

X : First face

Y : Abs diff

Conditional : $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						
	1	1	1	1	1	1



Examples

IIT Madras BSc Degree

DDTE

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$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1						
2						
3						
4						
5						
6						
	1	1	1	1	1	1



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X : First face

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Conditional: $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

x \ y	0	1	2	3	4	5
1	1/4	1/10	1/8	1/4	1/4	1/2
2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



Examples

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$\sum_x f_{X|Y}(x|y) = 1$

x \ y	0	1	2	3	4	5
1	1/4	1/10	1/8	1/4	1/4	1/2
2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



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Y : Abs diff

Conditional: $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1	1/4	1/10	1/8	1/4	1/4	1/2
2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



Examples

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DDTE

X : First face

Y : Abs diff

Conditional: $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1	1/4	1/10	1/8	1/4	1/4	1/2
2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



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Examples

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DDTE

X : First face

Y : Abs diff

Conditional: $f_{X|Y}(x|y)$

$\sum_x f_{X|Y}(x|y) = 1$

X \ Y	0	1	2	3	4	5
1	1/4	1/10	1/8	1/4	1/4	1/2
2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



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3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/4	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



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2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/6	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



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2	1/4	2/10	1/8	1/6	1/4	0
3	1/4	2/10	2/8	1/6	0	0
4	1/6	2/10	2/8	1/6	0	0
5	1/6	2/10	1/8	1/6	1/4	0
6	1/6	1/10	1/8	1/6	1/4	1/2
	1	1	1	1	1	1



Examples

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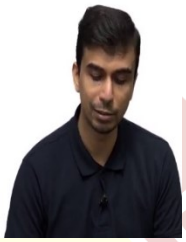
DDTE

X : First face

Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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DDTE

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Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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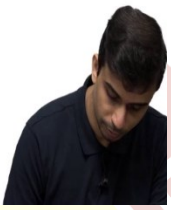
DDTE

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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X : First face

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Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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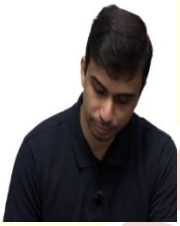
DDTE

X : First face

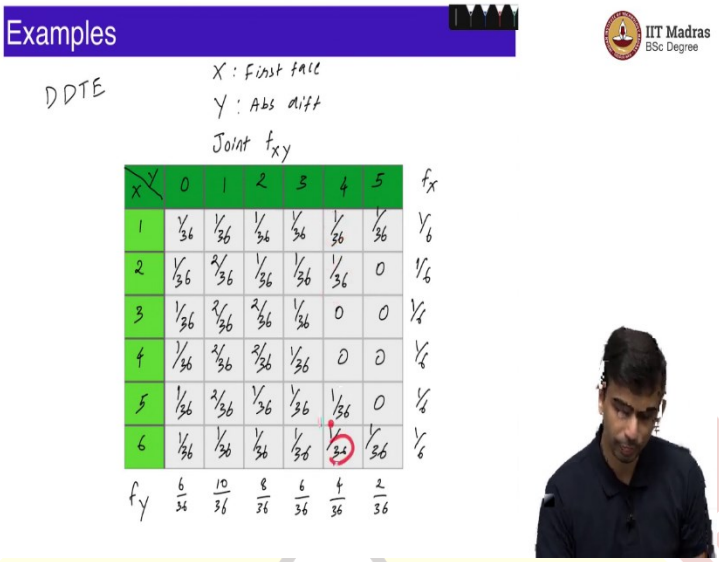
Y : Abs diff

Joint f_{XY}

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Now we will try to get the conditional distributions. So, the condition distributions, let us do f x given y. Let us have the same example, x being the first case y being absolute difference. We are now interested in finding the conditional distribution f x given y of x given.

What does this mean? This means that the probability of x equal to x given y equal to y, which means that if you sum over all x for any given y you should get 1. You sum over x f x given y of x given y should be equal to 1. Whatever be the value of y, y is given. So, you sum over all values of x, you, f, you should get back 1. So, that is the first clue.

And based on that we know that f x y f x given y of x given y is simply the joint divided by the marginal f y, which essentially corresponds to taking each column and dividing that by its sum. So, if you do that you get 1 by, take this column 1 and divide that by 6 by 36 you get 1 by 6.

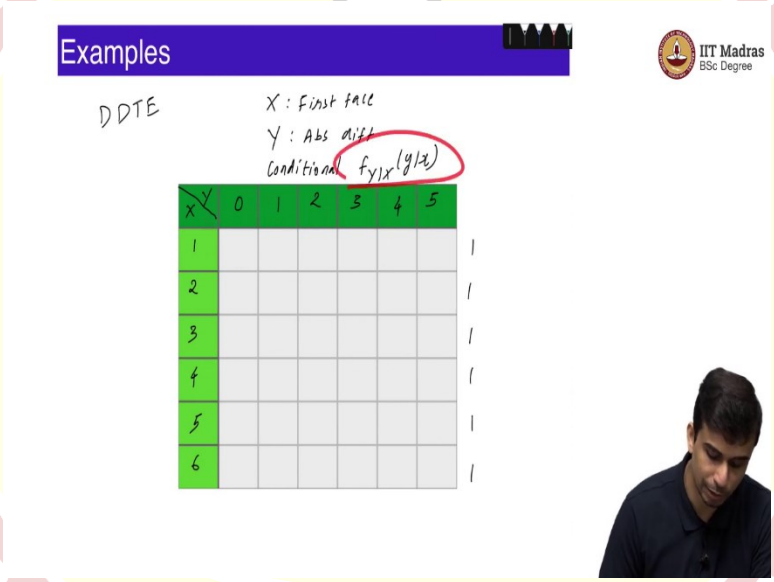
Let us take the second column and divide that by f y of y. So, this is, you get 1 by 10, 2 by 10, 2 by 10, 2 by 10, 2 by 10, and 1 by 10. Similarly, you can do that for the second, third column also. You get, sorry 1 by 8, 1 by 8, 2 by 8, 2 by 8, 1 by 8, 1 by 8. That is the 8 coming from, the 8 is coming from the column sum. So,rry.

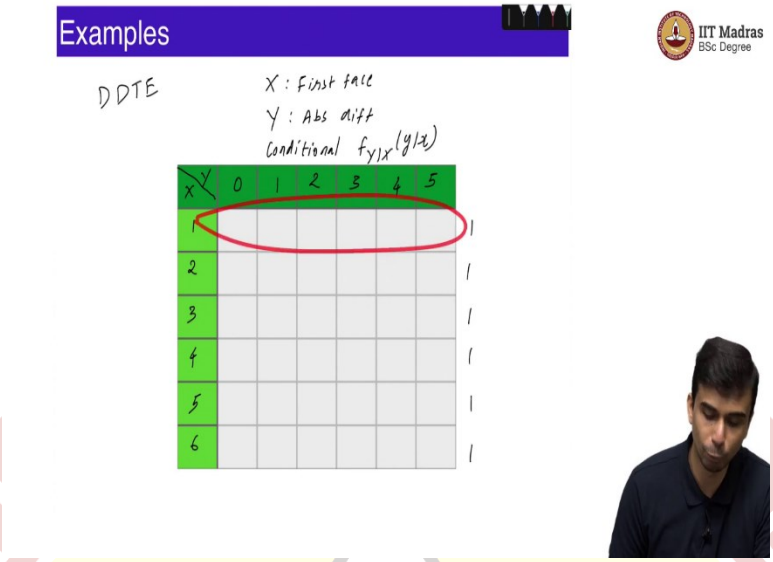
Similarly, you can do that for the fourth column. We get 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6. And the fifth column, we get 1 by 36. So, 1 by 4, 1 by 4, 0, 0, 1 by 4, 1 by 4. Finally, the last column you get half and half for.

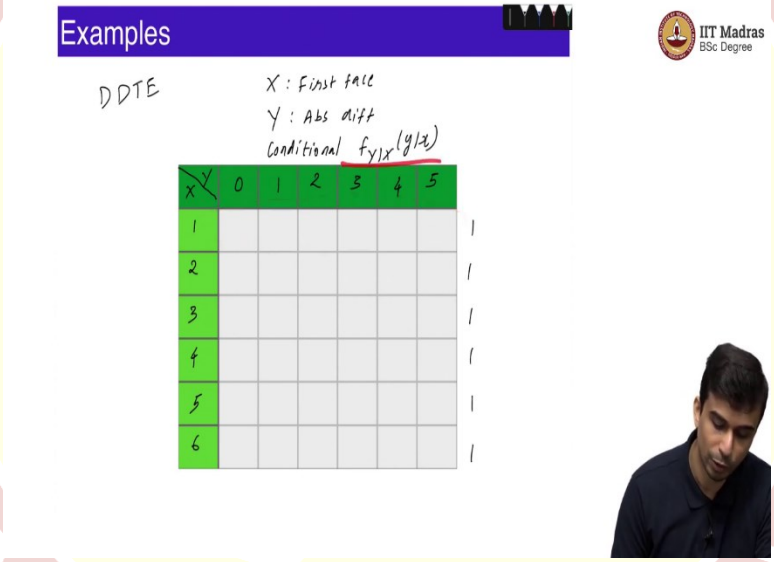
Let us try to interpret this. For example, let us try to interpret the last column. What does it say? It says that the probability of x is equal to 1, that is the formula first face being 1 is equal to half when the difference is 5. Or the probability of first phase being 6 when the difference is 5 is half. So, that means that when difference is 5, x, the first face is either 1 or 6 with equal probability. That makes sense to us.

So, when the difference between, when we know that the difference between the two, absolute difference between the two die throws is 5, well the first throw that should either be 1 or 6. That is the only two possibilities and both are equally likely. That is what this is capturing.

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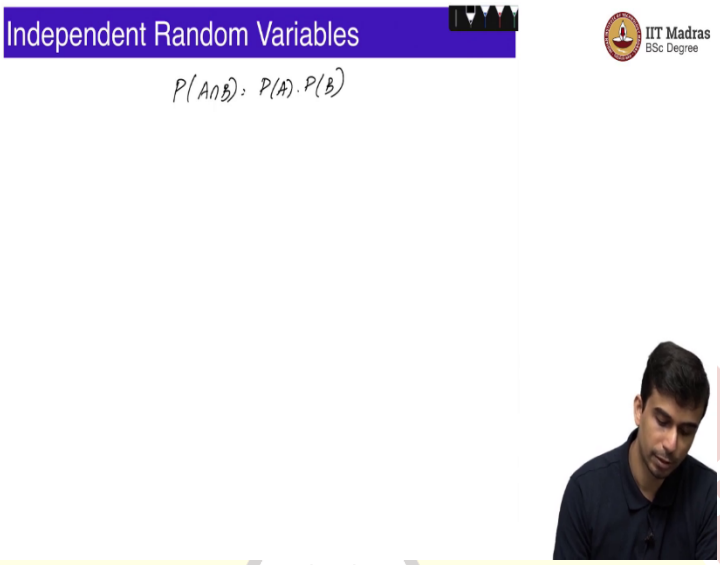


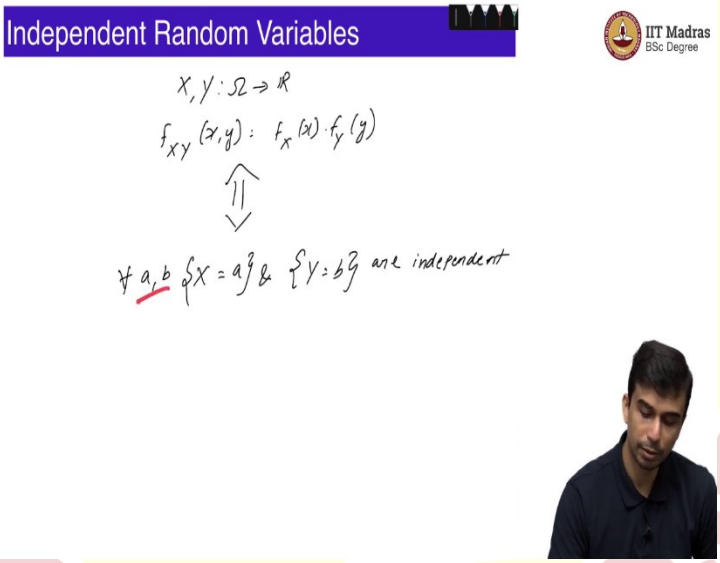




Similarly, each, and you can see that each column clearly sums to 1. So, this is the conditional distribution f x given y. And you can do a very similar thing for conditional f y given x. The only difference is that you would take the joint and take each row and divide that by each row. So, that way you get f y given x. You can interpret this in the exact same way. This is an exercise for you to do in your home.

(Refer Slide Time: 13:00)





Independent Random Variables

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$X, Y : \Omega \to \mathbb{R}$

$f_{XY}(x,y) : f_X(x) \cdot f_Y(y)$

$\Updownarrow$

$\forall\, a, b \; \{X = a\}$ & $\{Y = b\}$ are independent



Independent Random Variables

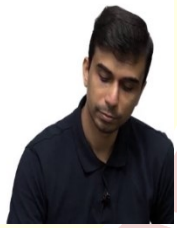
IIT Madras
BSc Degree

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$\forall\, a, b \quad \{X = a\}$ & $\{Y = b\}$ are independent

$\Updownarrow$

$\forall\, g, h \quad E[g(X)h(Y)] = E[g(X)] \cdot E[h(Y)]$



Independent Random Variables

IIT Madras
BSc Degree

$X, Y : \Omega \to \mathbb{R}$

$f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$

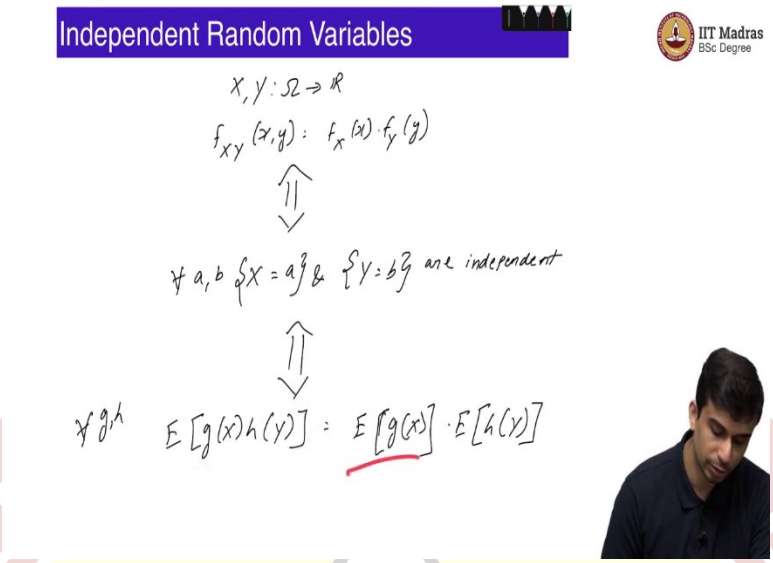
$\Updownarrow$

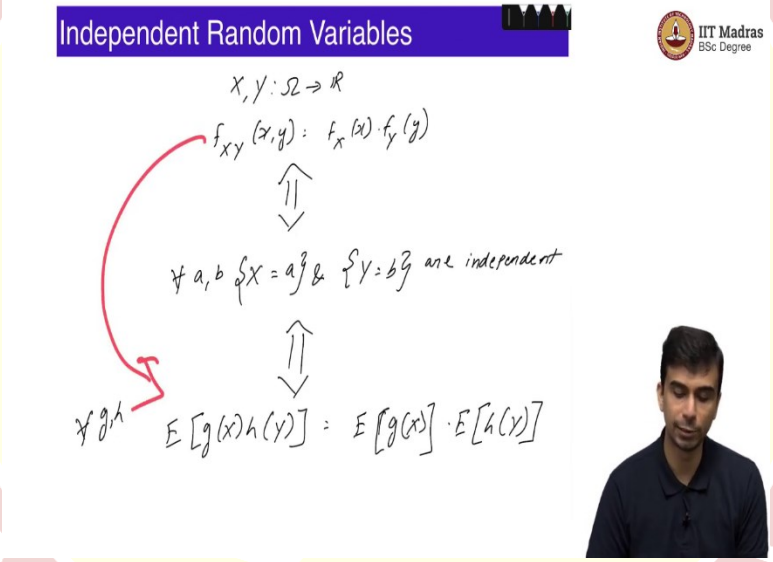
$\forall\, a, b \quad \{X = a\}$ & $\{Y = b\}$ are independent

$\Updownarrow$

$\forall\, g, h \quad E[g(X)h(Y)] = E[g(X)] \cdot E[h(Y)]$







Now we will move to another important concept, which is independent random variables. We have already seen independent events. We will say two events A and B are independent if probability of A intersection B is equal to probability of A into probability of B. So, that is the, we have, that is for independent events. But what does it mean to say two random variables are independent?

A very similar definition goes here, that is we will say two random variables x and y from ω to R are independent if the joint distribution f x y of x , y is equal to f x of x into f y of y. There are several equivalent ways to get this.

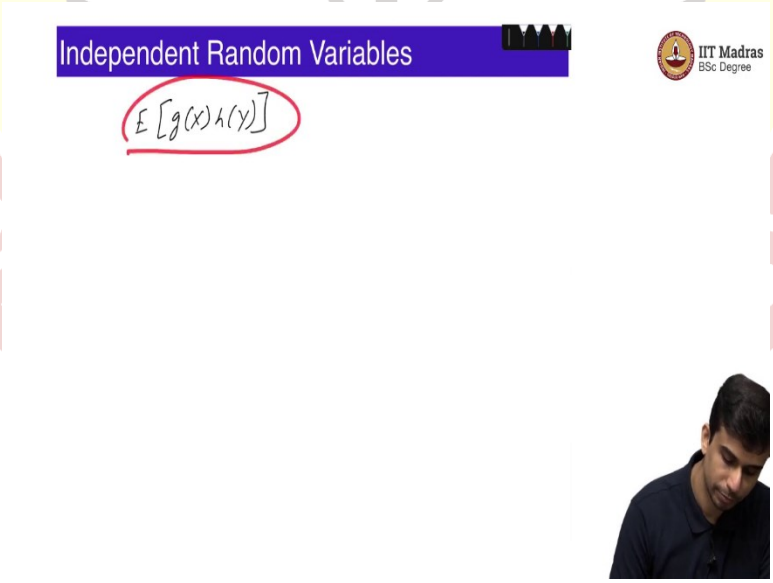
One is to say that for all values A and B, the events given by x equal to A and y equal to B are independent. Note that x and y are random variables, but for some constants A and B, if you say x equal to A, that is an event. So, independent, how are independent random variables and independent events related? Because we are using the same term here.

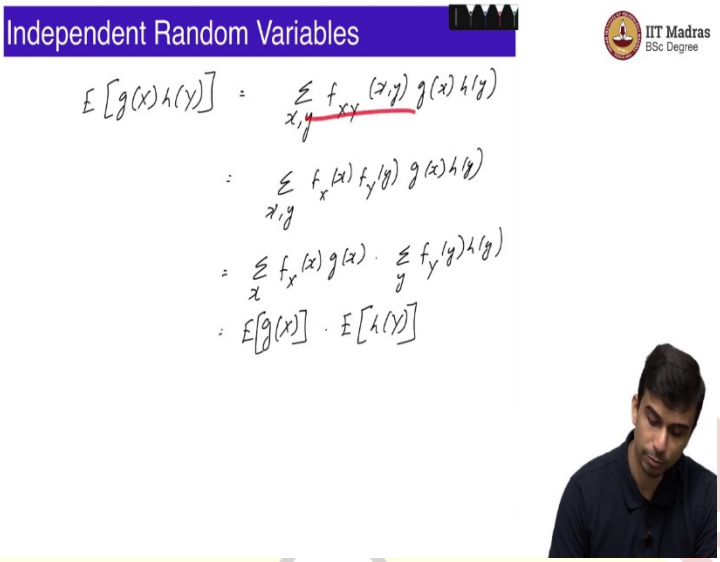
That is because you can convert one, you can define independent random variables in terms of independent events as follows. So, this, the equivalence between these two is pretty easy to show.

But there is also one other equivalent statement that you can say, which is that the expected value of f of x into g of y is equal to expected value of, let us not use f, let us use g and h, expected value of g of x into expected value of h of y. For, this is holds for all functions g and h. You take any function g and h, then the expected value of the product of g of x into h of y is simply expected value of g of x expected value of h of y.

So this is an equivalent definition for independence. We will actually prove from, you go from here to here you prove one direction. And the other direction is just as easy, actually. So, we will just prove this direction here.

(Refer Slide Time: 15:30)



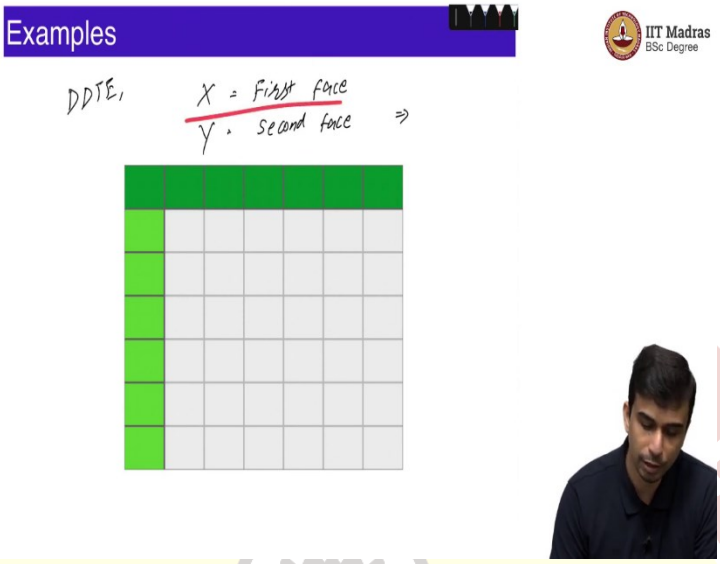


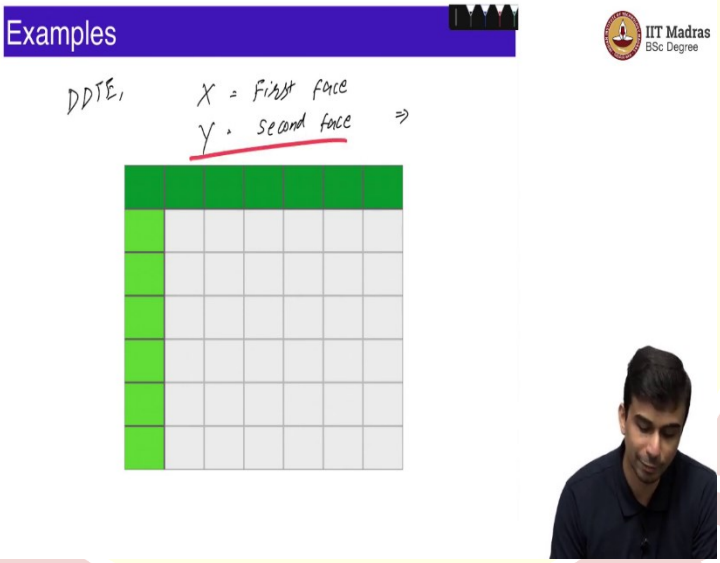
What is expected value of g of x into h of y? So, in this case this, there are two random variables x and y, so expected value of g of x into h of y is simply given by sum over x , y, f x y of x , y into g of x, sorry, into g of x into h of y. But we know that if the random variables x and y are independent, this is exactly equal to some over x , y, f x of x into f y of y into g of x into h of y.

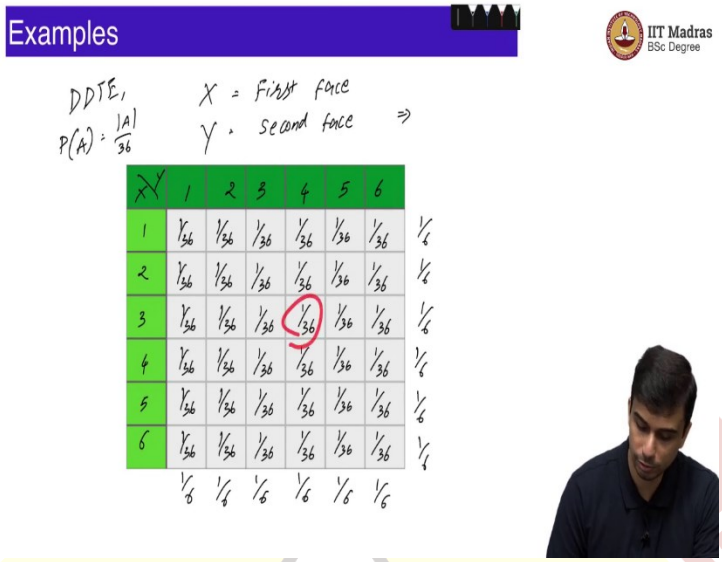
And by simple distributive law you can write this as sum over x f x of x into g of x into sum over y f y of y into h of y. This is exactly equal to expected value of g of x, and into expected value of h of y. So, this is a pretty easy proof and follows naturally from the definition of independent random variables.

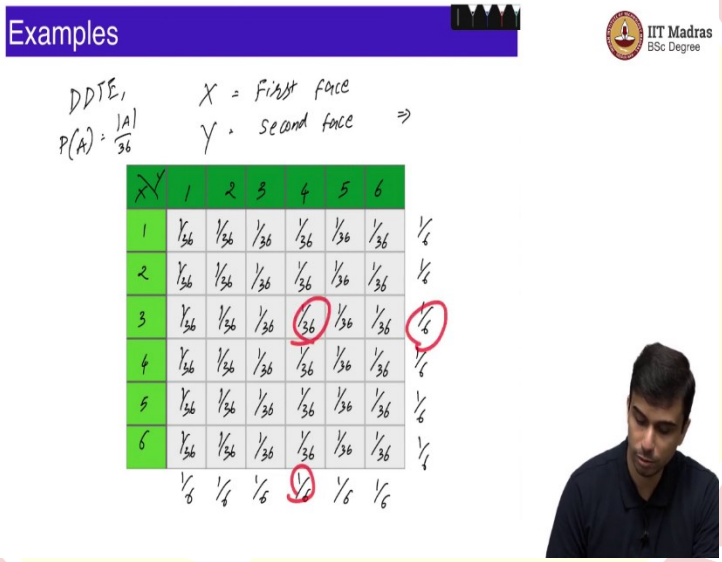
Independent random variables are actually 1 of the main, main bases from which you build more complicated dependencies. One useful interpretation is that if two random variables are independent, one random variable contains absolutely no information about the other. Knowing the value of one random variable gives you absolutely nothing about the other.

(Refer Slide Time: 17:25)









For example, the classic example would be, let us say you have two dice that you are throwing, the double dice through experiment with the standard probability measure counting probability measure, which corresponds to probability of any event A is cardinality of A by 36.

So, for that case where, which is equivalent to having two independent equal, two fair dice, let us say the first random variable is the, x is the first face, fir, the first die result and y is the second die result.

We can clearly see that these two are independent, actually it seems that they are independent random variables and that does reflect in the joint distribution. So, you can

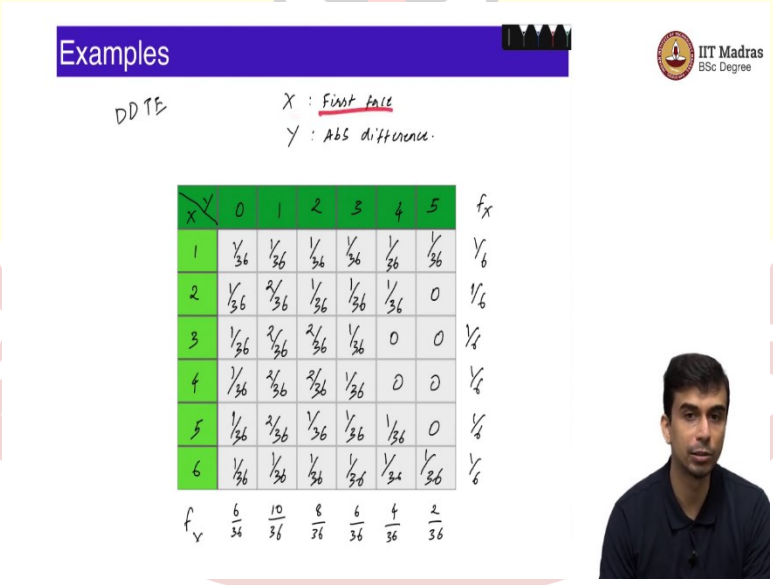
have, let us say x and y, x and y can take values 1, 2, 3, 4, 5, and 6. Similarly y also can take values 1, 2, 3, 4, 5 and 6.

So, we can fill in each cell in the entry here. What is the probability that x is equal to 1, and let us say y equal to 1? That is 1 by 36. In fact, the entire table is going to be filled with 1 by 36. This entire table is going to be filled with 1 by 36.

And you can compute the marginals of x and y by row sums and columns sums, which is 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6, 1 by 6. You can clearly see that to get any cell you can just multiply by the row sum and column sum, you get back that number. So, f x of x , y is exactly equal to f x of x into f y of y. So, this forms a pair of independent random variables. x , y are independent.

And the intuitive idea that x gives no information about y is obvious. So, if you know that the first die was 2, that does not help you in any way in finding out what the second die was.

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Examples

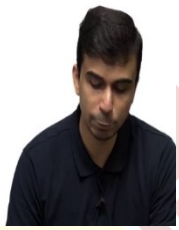
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DDTE

X : First face

Y : Abs difference.

X \ Y	0	1	2	3	4	5	f_X
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	2/36	1/36	1/36	1/36	0	1/6
3	1/36	2/36	2/36	1/36	0	0	1/6
4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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DDTE

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4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



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Examples

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DDTE

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4	1/36	2/36	2/36	1/36	0	0	1/6
5	1/36	2/36	1/36	1/36	1/36	0	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



Examples

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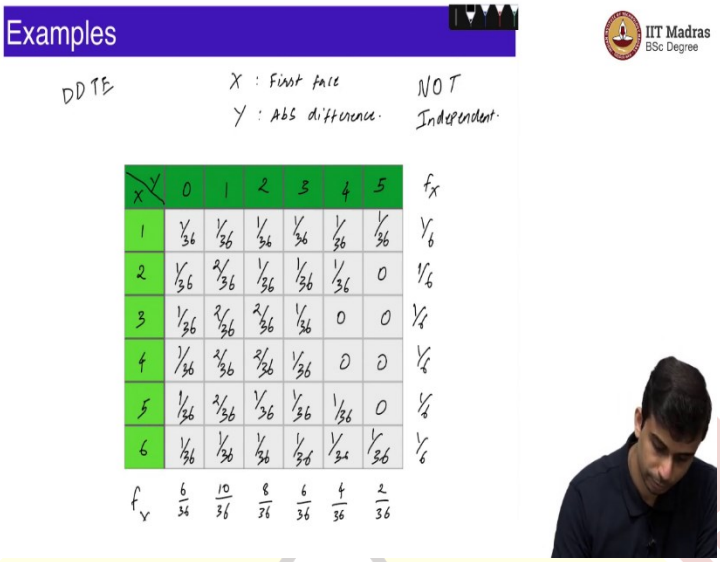
DDTE

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4	1/36	2/36	2/36	1/36	0	0	1/6
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6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
f_Y	6/36	10/36	8/36	6/36	4/36	2/36	



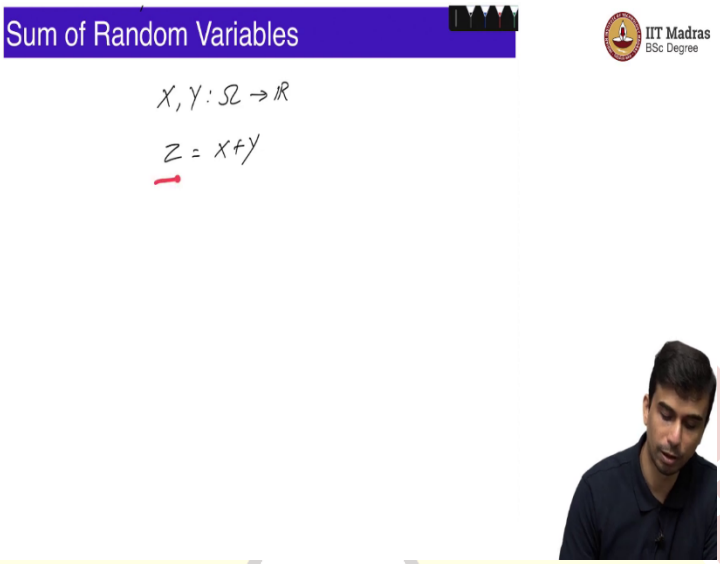


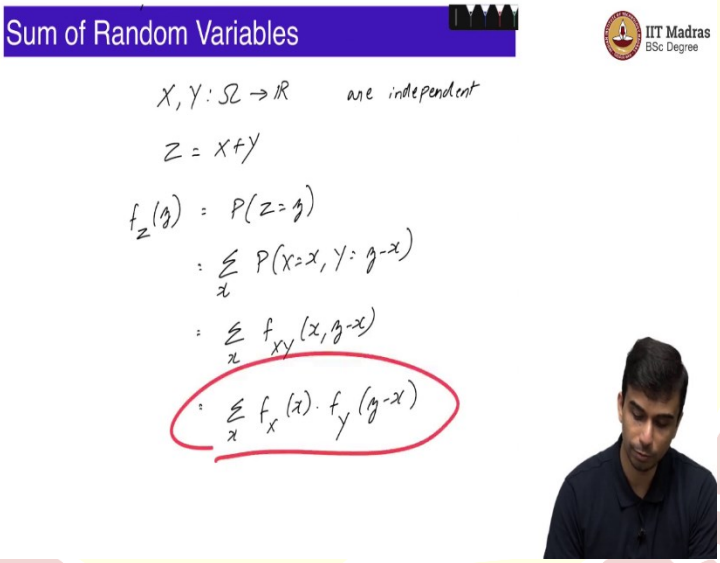
Here is another, here is the example that we did first. It is the same double dice throw experiment where x is the first face and y is the absolute difference. Here you can clearly see that f x of x , y is not equal to f x of x into f y of y.

For example, here you have 0. 0, 4 by 36 times 1 by 6 is clearly not 0. So, you have f x of x , y is not equal to f x of x into f y of y, and because x and y are not independent. Why is that? And how does x give, say, let us say some information about y.

Let us take the case where, let us say x is equal to 3. If you know x equal to 3, which is, the first face is 3, you know for certain that this, the difference cannot be 5. You know for certain. So, that is some information. Based on the knowledge of x, y can give you some information. You, mean, based on knowledge of x, you, you do get some information about y. You do get some information about y. So, that is the reason why x and y are not independent in this case.

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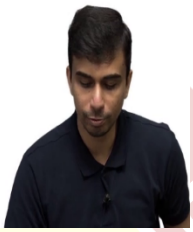
Sum of Random Variables

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$X, Y : \Omega \to \mathbb{R}$ are independent

$Z = X + Y$

$$f_Z(z) = P(Z = z)$$
$$= \sum_x P(X = x, Y = z - x)$$
$$= \sum_x f_{XY}(x, z - x)$$
$$= \sum_x f_X(x) \cdot f_Y(z - x)$$



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Sum of Random Variables

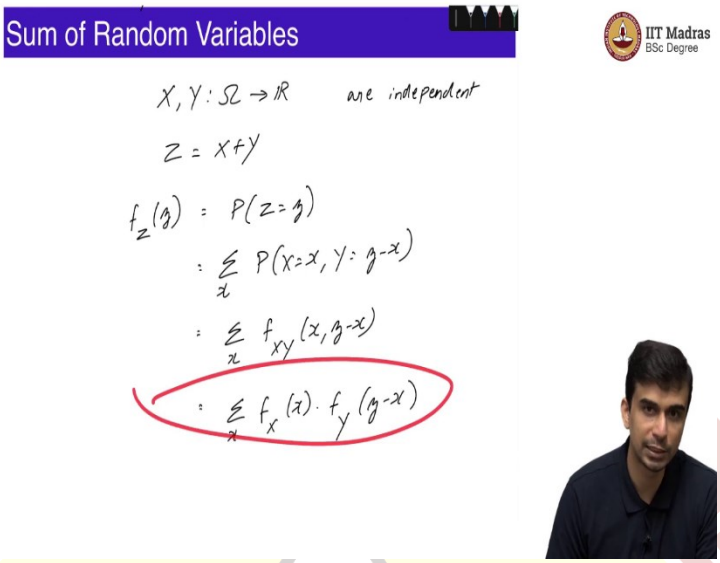
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$$f_Z(z) = P(Z = z)$$
$$= \sum_x P(X = x, Y = z - x)$$
$$= \sum_x f_{XY}(x, z - x)$$
$$= \sum_x f_X(x) \cdot f_Y(z - x)$$





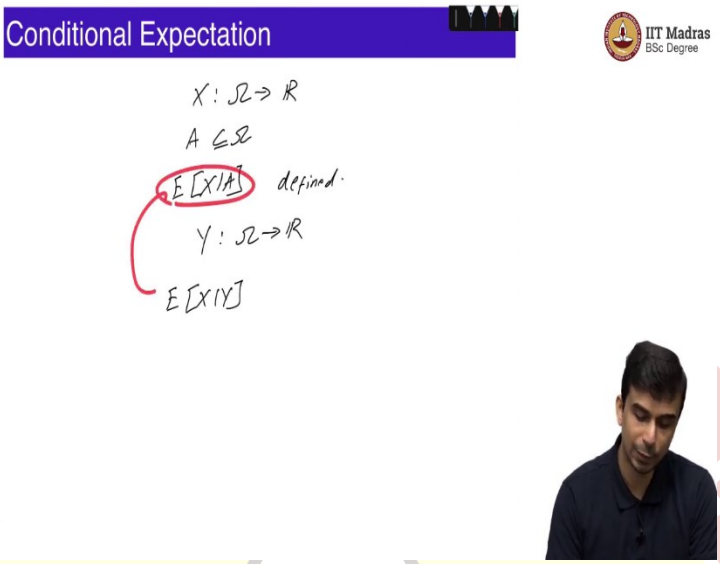
So, we will move on to the next topic, which is sum of independent random variables. So, we will not go into great detail about this topic. So, let us say you have two independent random variables x and y from ω to R. Let us say you have another variable z, which is x plus y. You want to know that, the probability mass function of z from the probability mass function of x and y.

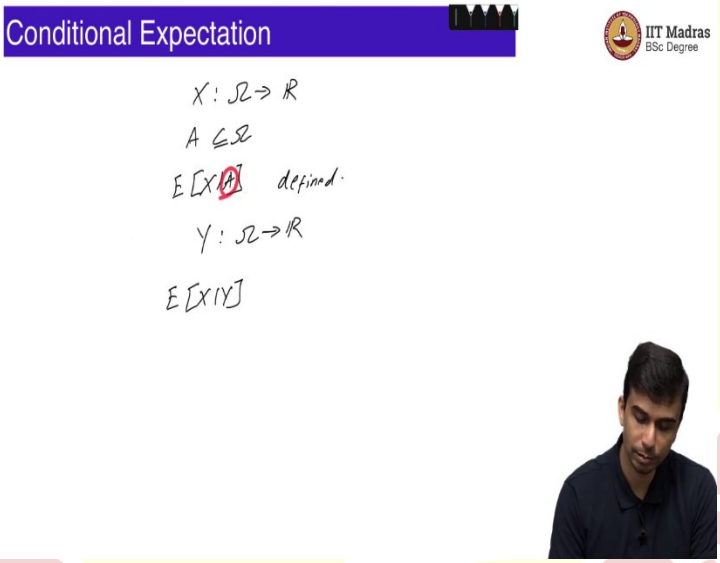
How do you go about doing that? Well, f z of z is equal to probability of z equal to z. When can z be equal to z? When, well, for any value of x when y equal to z minus x, you get. That so when probability of x equal to x and y equal to z minus x. And this is true for any value of x. So, you have to sum over x. So, this is some over x probability of x equal to x y, y equal to z minus x, which by our notation of the joint probability mass function, the sum over x f x y of x , z minus x.

And because we assumed x and y are independent, you can write f x y as f x into f y, so which means it will sum over x, f x of x into f y of z minus x. You can, this is called a convolution operation. And here is a simple exercise.

Let us take the double dice throw experiment where x is the first result of the first die throw, y is the result of the second die throw, and get, these two are clearly independent, use the convolution theorem to get the probability mass function of the sum of the two, two die throws. You know the answer anyway, already, but use the convolution theorem to get the answer here.

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Conditional Expectation

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$X : \Omega \to \mathbb{R}$

$A \subseteq \Omega$

$E[X|A]$ defined.

$Y : \Omega \to \mathbb{R}$

$E[X|Y]$



Conditional Expectation

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$X : \Omega \to \mathbb{R}$

$A \subseteq \Omega$

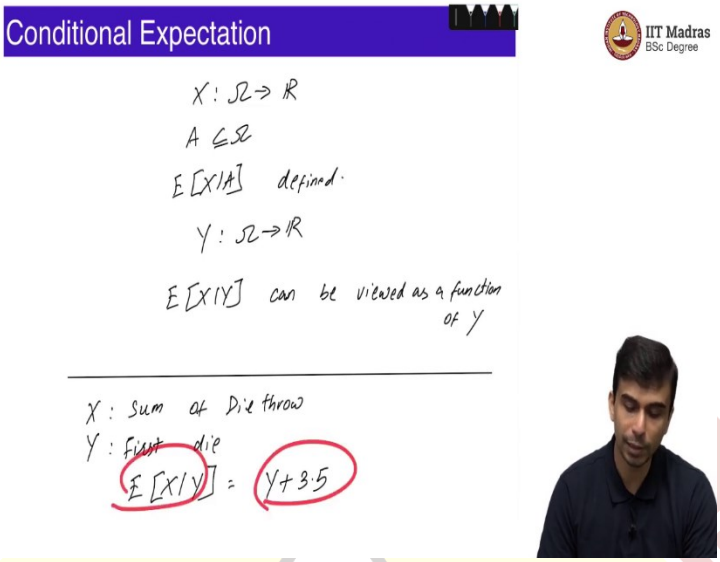
$E[X|A]$ defined.

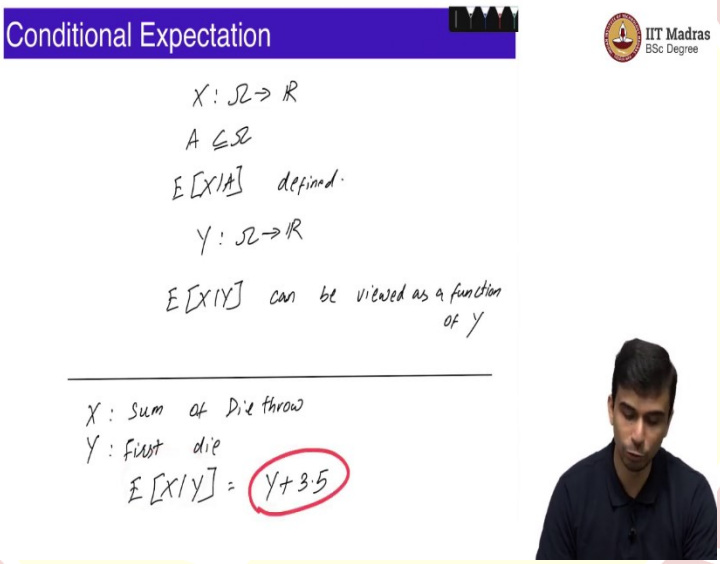
$Y : \Omega \to \mathbb{R}$

$E[X|Y]$



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We will move on to the next topic, which is conditional expectation. So, we have only defined conditional expectation conditioned on an event. So, we have a random variable x from ω to R, and let us say A is a subset of ω. We have already defined expected value of x given A. You have defined that.

But let us say you have another random variable y from ω to R. What does it mean to say expected value of x given y? First of all it you cannot directly use this statement because the expected value of f x given A is defined only for A being events. y is not an event, y is a random variable.

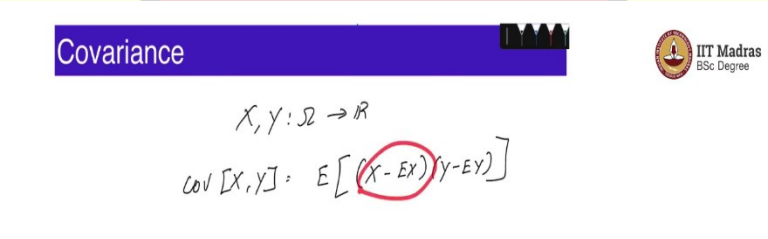
But there is always, the standard way to convert a random variable to an event is to assign a value to it. So, that is the, but if you assign different values to it, you get different events and based on that the answer, expected value of x given A is going to change. So, expected value of x given y can be viewed as a function of y, as a function of y. y is a random variable, but it can be viewed as a function of y.

Let us take a very simple example, and that should make a lot of things clear, that is, let us say x is the sum of die throws. Let us say, we have a double die throw experiment, sum of die throws. And y is the first die throw. What is the expected value of the sum, given the first die throw is y.

In this case, it is exactly equal to, mean, that means that you are taking the second, the first row as a constant. So, that means that expected value of x given y is exactly equal to y plus the expected value of the second die throw, which happens to be 3.5. You can see that expected value of x given y is a function of y. So, this is still a random variable but it is a function of y, that is the key thing to note here.

We will use the conditional expectation in this term in this form quite often in the next machine learning course. But right now, this is sufficient to know that conditional expectation, expected value of x given y is a function of the random variable.

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Covariance

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$X, Y : \Omega \to \mathbb{R}$

$Cov[X,Y] = E[(X - EX)(Y - EY)]$



Covariance

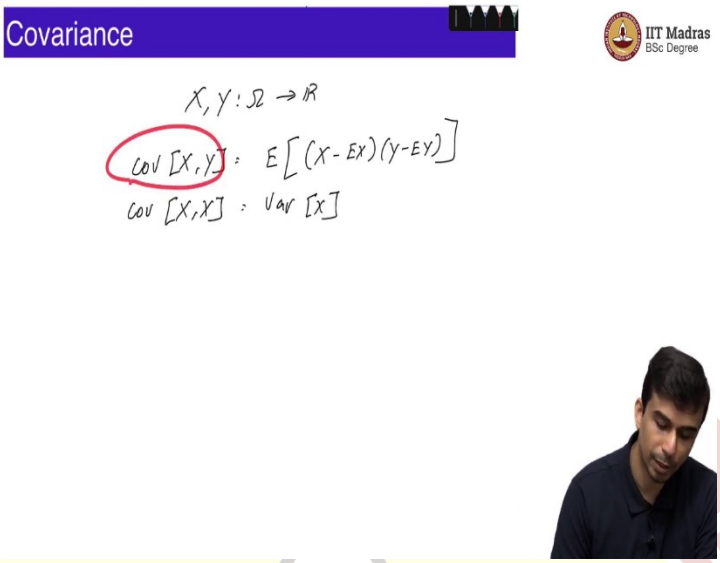
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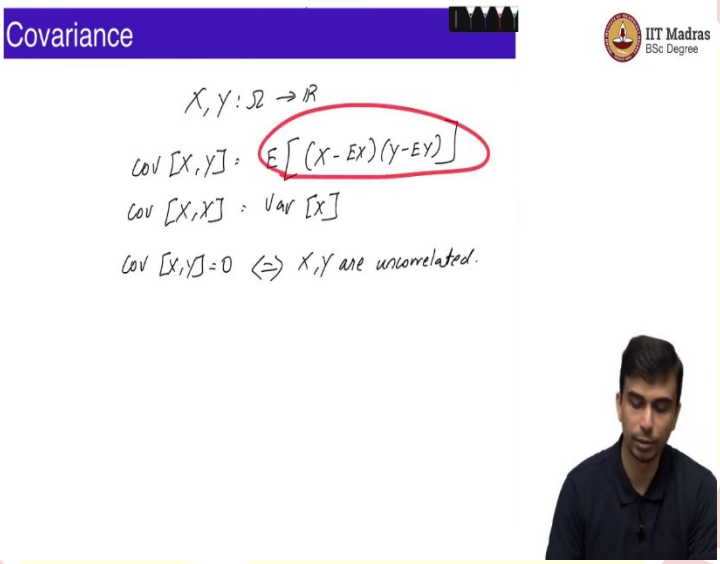
$X, Y : \Omega \to \mathbb{R}$

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Finally, we will move on to the next topic which is covariance. Let us say you have two random variables, x and y from ω to R. So, what is the covariance of the two random variables x and y? The formal definition of the covariance is the expected value of this particular object, x minus E x into y minus E y, x minus E x and y minus, this, this entire thing can be viewed as a random variable, so expectation of this random variable, that is, it is a perfectly valid way, definition here.

In particular, if you have covariance of x , x, you get back the variance of x. So, what is the interpretation of the covariance of a random variable. If you have the covariance of two

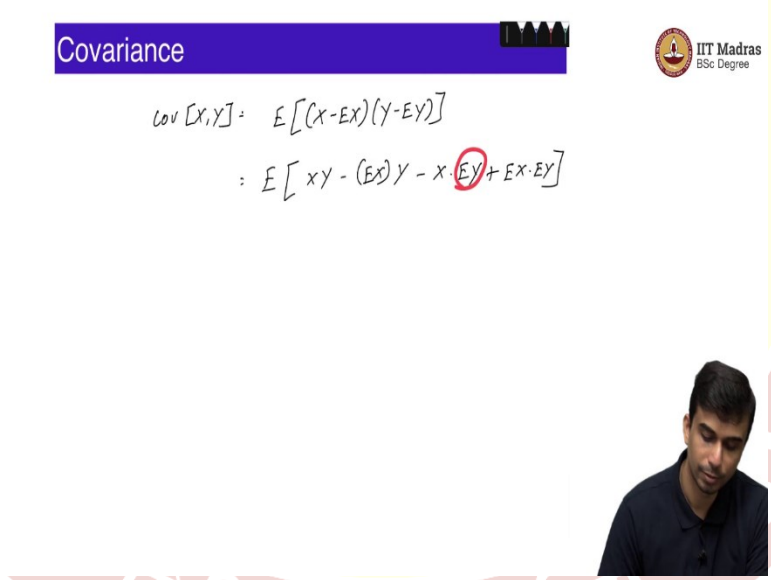
random variables as high and positive, that indicates that these two random variables generally go together. Either increase together or decrease together.

For example, you might hear, in some statistical studies saying that the covariance between heart risk and blood pressure is positive, which means that as the heart, as your blood pressure increases your heart risk also increases.

You can hear, it can also be negative, for example. You can say the covariance between your, let us say, exercise, number of hours you put into exercise and heart risk is negatively correlated. You can hear a statement like this, which means that as you exercise more your heart risk decreases. So, that means that it is a negative correlation.

Covariance can also be 0. And that special case is called uncorrelated. So, if covariance of x , y equal to zero, this particular pair of random variables are said to be uncorrelated.

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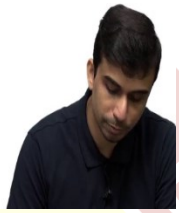
Covariance

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$$Cov[X,Y] = E[(X-EX)(Y-EY)]$$

$$= E[XY - (EX)Y - X \cdot EY + EX \cdot EY]$$

$$= E[XY] - EX$$



Covariance

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$$Cov[X,Y] = E[(X-EX)(Y-EY)]$$

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$$= E[XY] - EX \cdot EY - EX \cdot EY + EX \cdot EY$$

$$= E[XY] - EX \cdot EY$$



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Covariance

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If X, Y are Independent

$$E[XY] = EX \cdot EY \Rightarrow cov[X,Y] = 0$$



Covariance

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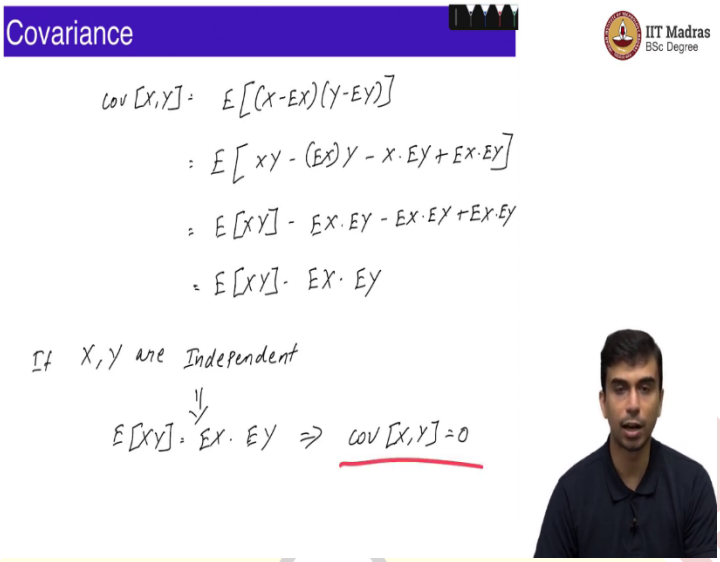
$$cov[X,Y] = E[(X-EX)(Y-EY)]$$
$$= E[XY - (EX)Y - X \cdot EY + EX \cdot EY]$$
$$= E[XY] - EX \cdot EY - EX \cdot EY + EX \cdot EY$$
$$= E[XY] - EX \cdot EY$$

If X, Y are Independent

$$E[XY] = EX \cdot EY \Rightarrow cov[X,Y] = 0$$



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And yeah, now let us actually redo this expectation and simplify this. Covariance of x , y is expected value of x minus E x into y minus E y. Simplifying this further, expected value of, we will just do the product inside, which is expected value of x y minus E x into y minus x into E y plus E x E y.

Ideally after every E, I should put a square brackets but I am taking it to be implicit. When I say E y, I mean E square brackets, within square brackets, y. Let us use the linearity of expectation to go further.

This is expected value of x y minus expected value of x, you can, E x is a constant, so you can take this out. So, this becomes E x into E y. Similarly, you have minus E x into E y plus E x into E y. Together you get E x y minus E x into E y. This would be another definition of covariance, you might see, and you can see that if x is equal to y you get back the old definition of variance, which is E x squared minus E x the whole squared.

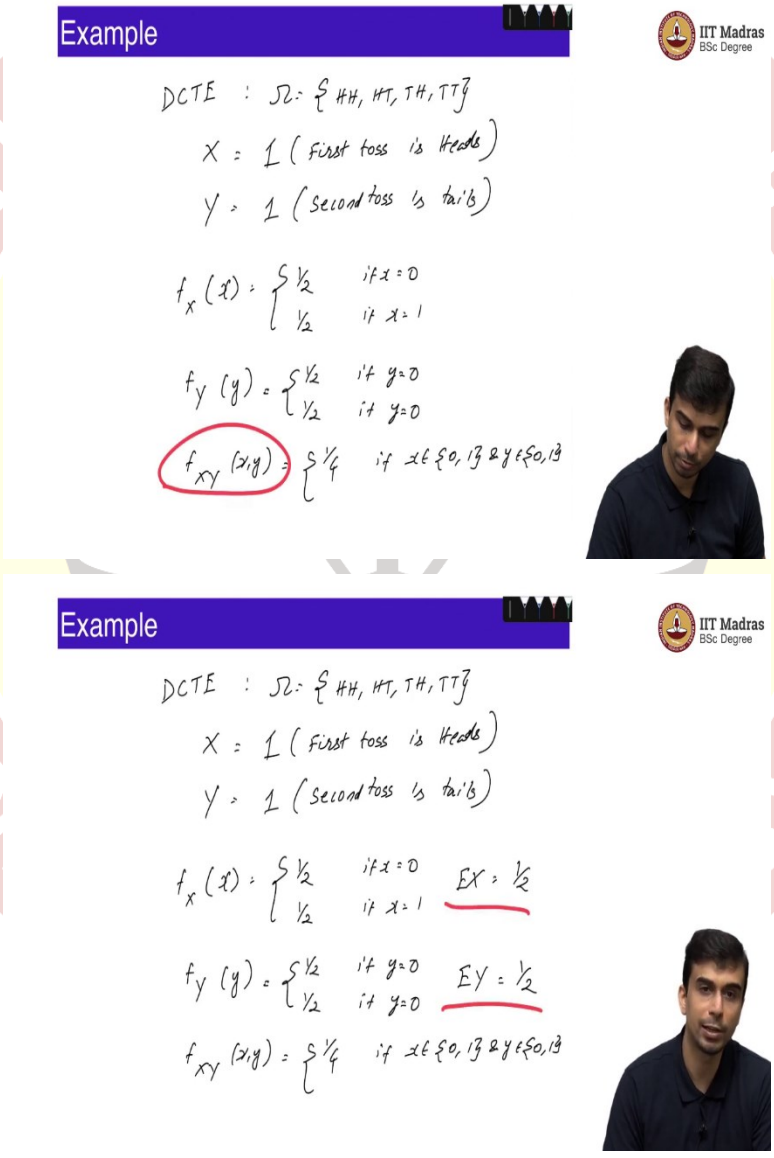
This should immediately tell you one thing, which is from our definition of independent random variables, we showed that if you have independent random variables x and y, expected value of product of, expected value of product is equal to the product of expectation.

So, which means that if x and y are independent, then this implies expected value of x y is equal to expected value of x into expected value of y, which is equal to saying that

covariance of x , y equal to 0. So, this is our first statement, that is, if x and y are independent then they are uncorrelated.

But this does not flow in this direction. You can have two random variables which are uncorrelated. Their covariance can be 0, but they can still be dependent. We will see a quick example of this.

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Before we see an example of that, let us see a simple example of something which we know to be independent. Let us take x to be the, let us take the double coin, coin toss experiment.

Let us take the double coin toss experiment, where your sample space ω is simply HH, HT, TH and TT.

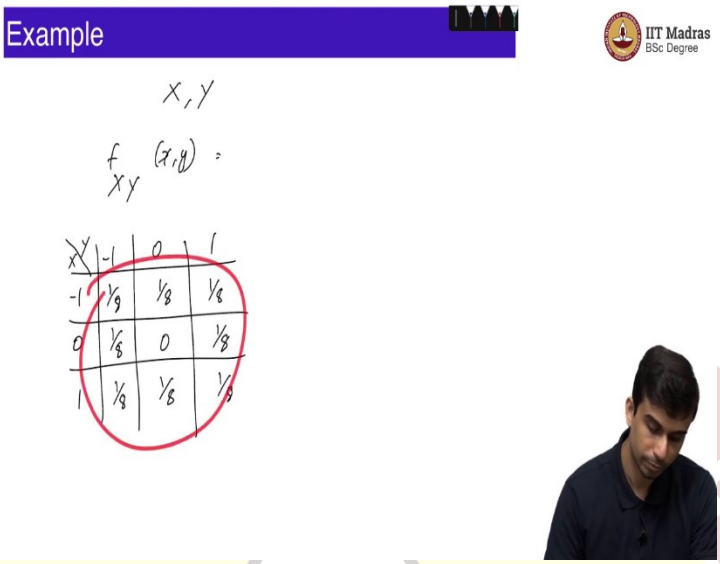
And let us take the two random variables, which are independent. Let us say x is equal to the indicator of first toss is heads, which means that, yeah, we know exactly what this means. Let us say y is the indicator of second toss is tails. x and y are independent. Why is that? Well, we can write down f x of x. f x of x is equal to half, if x equal to 0, half x equal to 1.

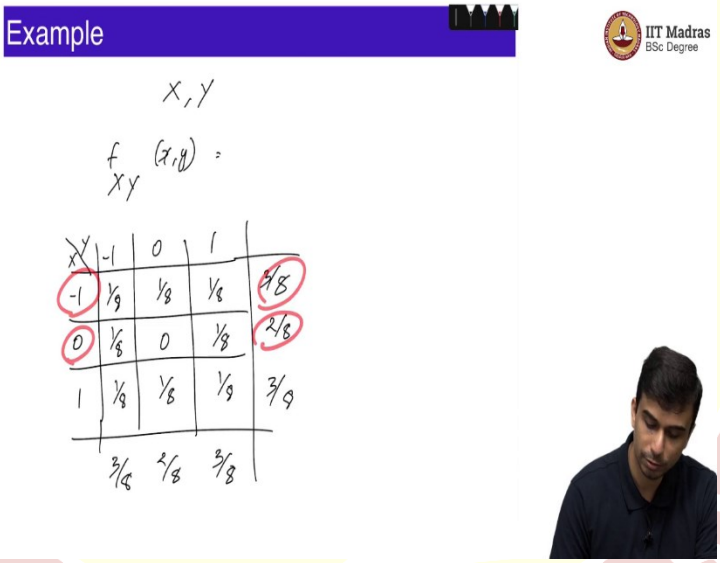
What is f y of y? Similarly, f y of y is also equal to half x equal to y equal to zero, half y equal to 1. What is the joint? Well, f x y of x , y equal to 1 by 4, if x equal to, I mean if x belongs to 0 , 1, our y belongs to 0 , 1. So, clearly you can see that f x of x into f y of y is equal to f x y of x , y. So, x and y are independent. So, in this case x and y also happen to be uncorrelated.

So, this is a simple example of pair of independent random variables, which are also encoded. Why is that? Well, what is E x, what is E y? E x is equal to half, and E y is also equal to half. What is E x y, well when is x y equal to 1. x y is equal to 1, when exactly when first kind toss is heads and the second tosses is tails. Every other situation is going to be equal to zero.

So, this, the probability of x y is equal to 1 is equal to 1 by 4, which means that expected value of x y is equal to 1 by 4, which is exactly equal to E x into E y. So, you have x and y are independent, and you also checked that they are uncorrelated.

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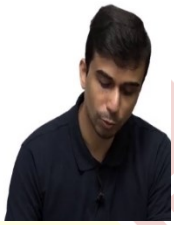
Example

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X, Y

$f_{XY}(x,y)$:

X \ Y	-1	0	1	
-1	1/8	1/8	1/8	3/8
0	1/8	0	1/8	2/8
1	1/8	1/8	1/8	3/8
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Example

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X, Y

$f_{XY}(x,y)$:

X \ Y	-1	0	1	
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0	1/8	0	1/8	2/8
1	1/8	1/8	1/8	3/8
	3/8	2/8	3/8	

$E[X] = 0$

$E[Y] = 0$

$E[XY]$



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$EX = 0$

$EY = 0$

$E[XY] = 0$

$P(XY = 1) = 1/4$

$P(XY = -1) = 1/4$



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Example

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X, Y

$f_{XY}(x,y)$:

$X \backslash Y$	-1	0	1	
-1	1/8	1/8	1/8	3/8
0	1/8	0	1/8	2/8
1	1/8	1/8	1/8	3/8
	3/8	2/8	3/8	

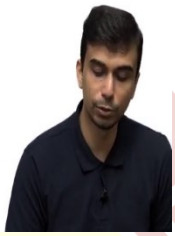
$EX = 0$

$EY = 0$

$E[XY] = 0$

$P(XY = 1) = 1/4$

$P(XY = -1) = 1/4$



Example

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$EX = 0$

$EY = 0$

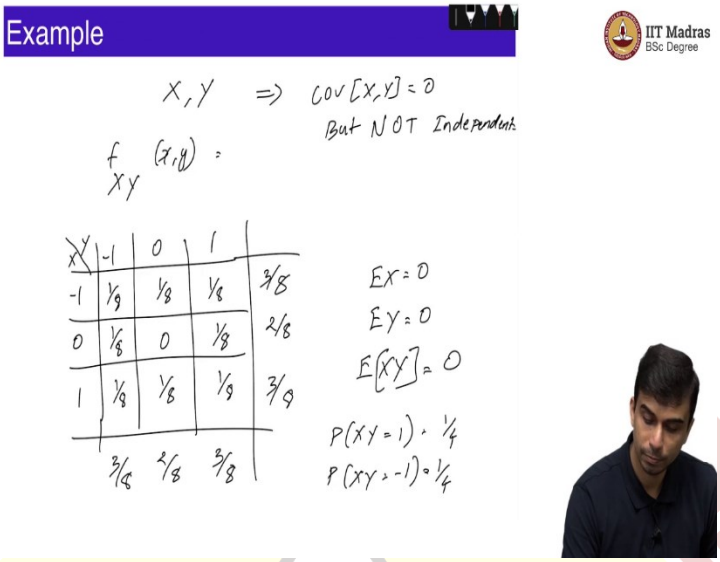
$E[XY] = 0$

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Let us take another example where you have a situation where we have uncorrelated random variables but they are actually dependent. Let us, let us say you have x and y. I will actually directly give the joint, x and y are defined as follows, which is f x y of x , y is given by this table. Let us, let us actually just directly give the table, say minus 1, 0, 1 minus 1, 0, 1. x and y take the values minus 1, 0 and 1. The joint distribution function is given by this table.

You can clearly see that this sums to 1. So, this is a valid probability mass, joint probability mass function. What is E x? Well, what we can compute the marginal probabilities f x and f y. So, if you did the marginal probabilities you would get 3 by 8, 2 by 8, 3 by 8. And similarly, you get 3 by 8, 2 by 8, 3 by 8.

So, what is E x, and what is E y? Well, E x is probability of x is equal to minus 1 is 3 by 8, probability of x equal to 0 is 2 by 8, probability of x equal to 1 is 3 by 8. So, clearly, E x equal to 0. Similarly, E y is also equal to 0.

What is E x y? x y can take values either 1, 0 or minus 1. So, x can take values minus 1, 0, 1, y can take values minus 1, 0, 1, so x y can also take values 1, 0 or minus 1. What is the probability that x y is equal to 1? Well, that can happen when x is equal to 1 and y equal to 1 or x equal to minus 1 and y equal to minus 1. So, which corresponds to these two numbers, which is minus 1, 1 by 8 plus 1 by 8, that is 1, by 4.

Similarly, probability of x y equal to 1 is equal to 1 by 4, probability of x y is equal to minus 1 is equal to 1 by 4, and probability of x y equal to 0 is half. So, clearly E x y is also equal to 0. So, E x equal to 0, E y equal to 0, and E x y is also equal to 0. So, this is clearly an uncorrelated pair of random variables.

But are they independent? You can clearly see they are not independent. Why is that? Because you have a 0 here. f x y of 0 , 0 is 0, but f x of 0 and f y of 0 are 2 by 8 and 2 by 8. So, if they were independent you would have had here 2 by 8 and 2 by 8, and not 0.

So even though these two random variables are not independent, they are uncorrelated. You can see that E x into E y is equal to E x y. So, the covariance of x and y happens to be 0. So, this is a pair of uncorrelated, covariance of x , y equal to 0, but not independent.

So, with that we wrap up this multiple, introduction to multiple random variables. In the next lecture we will take a look at specific example, named random variables, and their probability mass function, their cdf and some properties.